

Sourav Ray

Data Scientist | Predictive Modeling | Machine Learning | GenAI

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Professional Summary

Data scientist with 3 years of experience building Python and SQL data and ML systems on AWS, Snowflake and GCP, plus an M.Tech in Operations Research (NIT Durgapur) with a thesis on Transformer-based regression (R^2 up to 0.986). Strong in EDA, statistical evaluation, and pre/post-change validation, and in turning technical findings into business results: **65% lower Snowflake compute cost**, **60% faster incident detection (MTTD)** and **65% faster resolution (MTTR)** using LLM agents. Experienced in documenting work clearly and owning deliverables end to end with stakeholders.

Technical Skills

Programming & Analysis: Python (pandas, NumPy, scikit-learn), SQL (advanced, large-scale), Shell Scripting
Statistics & ML: Regression and classification, EDA, outlier detection (Z-score/IQR), time-series forecasting, hypothesis testing (t-test, p-value), MLE, PCA, mathematical optimization, model evaluation (RMSE, MAE, MAPE, R^2)
Deep Learning: TensorFlow, Keras, Transformers (self-attention), LSTM/GRU/Bi-LSTM, 1D CNN
GenAI & LLMs: LLM agents, RAG, Model Context Protocol (MCP), multi-agent orchestration, prompt engineering, Snowflake Cortex, hybrid retrieval and reranking
Cloud & Data: GCP (BigQuery, Vertex AI), Snowflake (Snowpark, Snowpipe), AWS (S3, Lambda, Kinesis, DMS), DBT, Airflow, Kafka, CDC, data contracts and data quality
Tools & DevOps: Git, GitHub Actions, Docker, Kubernetes, Terraform/CDKTF

Professional Experience

Factspan Analytics

Senior Analyst

Bengaluru, India

Sep 2025 – Present

Snowflake Cost Optimization Agent

May 2026 – Present

- Cut Snowflake compute cost by **65%** across 60 DBT models and 70 virtual warehouses by profiling queries (disk spillage, Cartesian joins) and tuning warehouse auto-suspend and right-sizing.
- Built an autonomous LLM agent (Snowflake Cortex, Snowpark, DBT, MCP) that diagnoses cost anomalies and automates fix recommendations via GitHub pull requests and Jira tickets as Lead Engineer.
- Designed a three-part validation suite (row count, column count, MD5 hash diffing) run pre- and post-change, with post-deployment telemetry to verify savings without data drift.
- Automated RBAC and account-level parameter management across dev, staging, and production using Terraform and staged rollouts.

AI-Powered Network Incident Orchestrator

Jan 2026 – Present

- Reduced MTTD by **60%** and MTTR by **65%** using a multi-agent AI system: one orchestrator directing five specialist sub-orchestrators for automated troubleshooting and diagnostics.
- Correlated incidents, telemetry, topology, and change history across 20+ enterprise tools (ServiceNow, Datadog, Splunk, NetBox, PagerDuty) into a single LLM-generated root-cause summary.
- Built and deployed Python MCP servers on Kubernetes with workload autoscaling; integrated automated incident summaries into ServiceNow and Microsoft Teams bridges.

Factspan Analytics

Analyst

Bengaluru, India

Sep 2023 – Aug 2025

Customer 360 Data Mesh & Governance Platform

Sep 2023 – Dec 2025

- Architected Bronze/Silver/Gold lakehouse layers across 100–200 DBT/SQL models for a petabyte-scale Customer 360 platform integrating 1,000+ sources across five patterns.
- Improved data accessibility by **60%** through governed data products, multi-stage reconciliation, and contract-driven onboarding.
- Cut data modeling effort by **80%** via metadata-driven code generation, automating data contract creation for 100+ tables in 10 minutes.
- Ingested 500 tables from Oracle, MySQL, MariaDB, and PostgreSQL via AWS DMS (full load + CDC) with zero data loss using LSN handoffs and automated Snowpipe streaming.
- Accelerated CI/CD delivery with Terraform/CDKTF and GitHub Actions, deploying ephemeral per-PR test environments to validate changes before merge.

Projects

Job Discovery & Application Assistant – Retrieval-First AI Engine (Independent, GitHub)

- Built a high-recall hybrid retrieval engine (Dense embeddings + BM25) with Reciprocal Rank Fusion (RRF), Cross-Encoder reranking, and MMR diversification (Top-100 → Top-20 → Top-10–15).
- Developed an offline IR evaluation suite (Recall@K, Precision@10, NDCG@K, MRR), candidate-job fit analysis, and an interactive Streamlit dashboard scheduled daily via GitHub Actions.

Spatio-Temporal Air Pollution Prediction using a Transformer (M.Tech Thesis)

May 2023

- Designed an encoder-only Transformer with masked-value pre-training to forecast PM2.5, PM10, SO₂, and NO₂ on Beijing AQI data (TensorFlow, Keras, scikit-learn).
- Outperformed GRU, LSTM, Bi-LSTM, and CNN baselines with highest R² across all 4 pollutants (up to 0.986 on NO₂), confirmed statistically by residual t-tests.
- Built full data pipeline: outlier detection (Z-score/IQR/Tukey), rolling windows, normalization, and Pearson spatial-correlation analysis across 12 monitoring stations.

Education

National Institute of Technology (NIT), Durgapur

2021 – 2023

M.Tech, Operations Research | CGPA 8.36/10

Government College of Engineering and Textile Technology

2015 – 2019

B.Tech, Textile Technology | CGPA 7.43/10

Certifications

Google Cloud Professional Data Engineer | Snowflake SnowPro Advanced Data Scientist | AWS Certified Machine Learning – Specialty | Snowflake SnowPro Advanced Architect | AWS Certified Solutions Architect – Associate